MIL-T-18881(NOrd)

30 June 1955
SUPERSEDING
NAVORD OS 5896
7 April 1950

MILITARY SPECIFICATION

TEST SET MK 171 MOD 0

1. SCOPE

1.1 This specification governs the manufacture, assembly, and preparation for delivery of the Test Set Mk 171 Mod 0 and the methods of inspection and tests upon which acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621 -- Boxes; Wood, Nailed and Lock-Corner.
PPP-B-601 -- Boxes; Wood, Cleated-Plywood.

MILITARY

JAN-P-106 -- Packaging and Packing for Overseas Shipment-Boxes; Wood, Nailed.
MIL-P-116 -- Preservation; Methods of.
MIL-V-173 -- Varnish, Moisture and Fungus Resistant, for the treatment of communications, electronic, and associated electrical equipment.

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STANDARDS

MILITARY

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 130089 -- Test Set Mk 171 Mod 0,
486767 -- Spring,
and all documents listed thereon.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Test Set Mk 171 Mod 0 covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings No. 130089 and all documents listed thereon.

3.1.1 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Moisture - and fungus-resistant treatment. -- Materials and parts of this test set which are not moisture and fungus resistant shall be treated, before assembly, with a varnish conforming to Military Specification MIL-V-173.

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3.3 Dummy pistol. -- A dummy pistol shall simulate the weight, approximate center of gravity, snap ring flange, overall diameter and width of the base of a Depth Charge Pistol Mk 15 Mod 0. The snap ring flange shall be mounted in the same relative position as on the Depth Charge Pistol Mk 15 Mod 0. Application may be made to the Government for the use of a Depth Charge Pistol Mk 15 Mod 0 for this test.

3.4 Arming drag mechanism operation. -- The mechanism shall operate to rotate a Depth Charge Pistol Mk 15 Mod 0 (or Dummy Pistol as specified in 3.3) smoothly at five to ten revolutions per minute (rpm), without perceptible acceleration or deceleration, when mounted on the roller assembly, but not connected to the coupling of the mechanism.

3.4.1 Bearing assembly. -- The bearing assembly shall be aligned to receive the snap ring flange of a Depth Charge Pistol Mk 15 Mod 0 (or Dummy Pistol as specified in 3.3) when the pistol is mounted on the roller assembly without cocking or exerting a cantilever load to the pistol shaft. The bearing assembly shall rotate freely without sticking on high spots (with the coupling mounted without the rocker) when a torque of two inch-ounce is applied to the coupling by accepted standard methods.

3.4.2 Spring adjustment. -- The tension of the spring (Bureau of Ordnance Drawing No. 486767) shall be adjusted to permit closure of switch S1 and lighting lamp PL2, when a 13 ± 1 inch-ounce torque is applied to the coupling with the calibrated torque tool and with the test set in a horizontal position. After adjustment, the adjusting nut shall be secured by applying glyptal.

3.4.3 Roller assembly. -- The rollers shall be parallel and aligned to rotate freely without binding or sticking on high spots.

3.4.4 Lamps PL2 and PL3. -- With switch S3 held closed, lamp PL3 (green) shall light and when switch S1 is actuated, lamp PL2 (red) shall light.

3.5 Spin functioning mechanism operation. -- The spin functioning mechanism shall rotate the snap ring flange coupling at 5,500 ± 500 rpm within 1 1/2 seconds after switch S3 is closed, in such a manner that avoids excessive vibration of all the rotating parts.

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3.5.1 Bearing assembly. -- The bearing assembly shall be aligned to receive the snap ring flange of a Depth Charge Pistol Mk 15 Mod 0 (or Dummy Pistol as specified in 3.3) when the pistol is mounted on the cradle, without cocking or exerting cantilever load to the pistol shaft. The bearing assembly shall rotate freely without binding or sticking on high spots (tested without the actuator switch and motor belt assembled) when a torque of 4 inch-ounces is applied to the coupling by accepted standard methods.

3.5.2 Actuator Switch. -- The actuator switch shall be mounted to cut off the motor and the timer at an 1/8 ± 1/16 inch deflection of the coupling.

3.5.3 Timer. -- The timer shall start when switch S3 is held closed and shall stop after switch S2 is opened. The timer shall accurately measure a time interval of two seconds throughout its entire range within ± 0.03 seconds.

3.5.4 Lamp PL1. -- With switch S3 held closed, lamp PL1 shall light only when pins four and eight of cable plug assembly are shorted.

3.6 Covers and catches. -- Covers and catches shall function in a manner intended. Covers shall fit in a manner to keep out insects and foreign matter and shall open and close without jamming or interference. Catches shall operate to secure covers, when closed or open for operation of the test set, without jamming or interference.

3.7 Vibration. -- After being subjected to the vibration schedule specified in paragraph 4.4.3 this test shall meet all other requirements of this specification.

3.8 Cleanliness. -- Necessary precautions shall be taken to protect test set parts and assembly from moisture, solder splashes, dirt or foreign matter during manufacture and assembly.

4. QUALITY ASSURANCE PROVISION AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the

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acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Military Standard MIL-STD-105, "Sampling Procedures and Tables for Inspection by Attributes".

4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of one Test Set Mk 171 Mod 0 shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These test sets shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

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4.4 Tests.

4.4.1 Arming drag mechanism operation. -- Rotation may be checked by using a chalk mark on the pistol and a stop watch, or any accepted standard method. Any perceptible deceleration or acceleration of the mark in its rotation shall be cause for rejection.

4.4.2 Spin functioning mechanism operation. -- An oscilloscope shall be used to indicate the rate of change of current consumption of the motor and recorded on a film for accurate reading. Other reliable methods which are accurate to within .05 seconds may be used for reading the time for the motor to come up to speed. Failure to rotate the coupling to the minimum speed within 1-1/2 seconds shall be cause for rejection.

4.4.3 Vibration Test. -- The test specified in this section will be performed by an agency designated by the Bureau of Ordnance on the preproduction sample and any other samples as may be specifically designated by the Bureau of Ordnance.

4.4.3.1 Method of mounting. -- The test set shall be rigidly mounted upright to the vibration table. The clamping to be such that the test set is not distorted.

4.4.3.2 Direction of vibration. -- The test set shall first be vibrated in a direction perpendicular to its base and then in a direction parallel to the base of the test set.

4.4.3.3 Type of motion. -- The motion of the vibration table shall be simple harmonic.

4.4.3.4 Frequency range. -- The frequency shall be increased from 700 cycles per minute to 3000 cycles per minute in steps of 100 cycles per minute.

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4.4.3.5 Amplitude. -- The amplitude (single) shall be varied with the frequency shown in Table I.

Table I. - Amplitude Schedule

Frequency Range (Cycles per minute)	Peak Amplitude (Inches)
700 - 1500	0.030 ± 0.002
1600 - 2000	0.020 ± 0.002
2100 - 2500	0.013 ± 0.001
2600 - 3000	0.009 ± 0.001

4.4.3.6 Duration. -- The test set shall be vibrated continuously for 12 hours in each direction specified in 4.4.3.2, one-half hour at each frequency.

4.4.4 Component parts. -- The component parts shall be tested for compliance with the requirements of 3.2 through 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Domestic shipment. -- Each test set shall be packaged in a Style RSC corrugated fiberboard box fabricated from double-face board (275 pound minimum bursting strength) conforming to the requirements of Federal Specification LLL-B-631. Sufficient fillers of the same material shall be used inside this container to provide a tight pack and to prevent direct contact between protrusions of the test set box and the walls of the container.

5.1.2 Overseas shipment. -- Each test set, packaged as described above, shall be further packaged Method IIb, less desiccant, conforming to the requirements of Military Specification MIL-P-116.

5.2 Packing.

5.2.1 Domestic shipment. -- each test set, packaged as specified in 5.1.1, shall be packed in a wood-cleated plywood or nailed wood box conforming to the requirements of Federal Specifications PPP-B-601 or NN-B-621, respectively.

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5.2.2 Overseas shipment. -- Each test set, packaged as specified in 5.1.2, shall be packed in a wood-cleated plywood or Style 2, 2-1/2, 3, or 4 nailed wood box conforming to the requirements Military Specification JAN-P-106 or Federal Specification PPP-B-601, respectively. When nailed wood boxes are used, the lumber shall be surfaced on both sides.

5.3 Marking. -- In addition to any special markings required by the contract or order, all markings shall be in accordance with Military Standard MIL-STD-129, "Marking for Shipment and Storage". The following marking shall appear on the top, one (1) end and one (1) side of the shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

6. NOTES

6.1 Intended use. -- The Test Set Mk 171 Mod 0 is used to test the air-arming mechanism of Depth Charge Pistol Mk 15 Mod 0 during acceptance, assembly and issue of the Depth Charge Pistol.

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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